Positive Lyness Dynamics: Complete Ovals, Abel Rotations and Return Matrices

HCS-C390 source-theorem and reproducibility package

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Abstract

We give a complete positive-real orbit description of the autonomous Lyness map for every positive parameter. Strict convexity of its logarithmic integral proves that every noncentral two-sided orbit lies on one compact regular oval, with no missing positive components or escape trajectories. A specified clockwise section gives the map's rotation number as a ratio of two Abel integrals with the same quartic square root. This fixes the orientation and distinguishes auxiliary Hamiltonian time from one discrete iterate. Rational rotations classify every least period, every repetition and all primitive orbit families; irrational rotations give dense orbits on their own oval. The return derivative is an exact unipotent shear, without an unsupported claim that its shear coefficient never vanishes. The unique center, the globally five-periodic parameter and the coordinate boundary are treated separately. These are classical source-local results reconstructed with explicit proof and independently checkable identities, not a new-literature-priority claim or a target determinant construction.

中文摘要

本文给出自主莱尼斯映射在全部正参数下的完整正实轨道描述。
对数坐标中的严格凸性证明每个非中心双向轨道均位于唯一紧正则卵形线上， 不存在遗漏的正分支或逃逸轨道。
固定的顺时针截面将旋转数写为同一四次根式下两段阿贝尔积分之比， 并明确区分辅助哈密顿时间与离散迭代时钟。
有理旋转确定最小周期、重复与全部本原轨道族； 无理旋转给出本卵形线上的稠密轨道。
返回导数是精确幂零剪切，但本文不假定剪切系数处处非零。 唯一中心、全局五周期参数与坐标边界均单独处理。
这是具有明确经典归属的源系统重建，不声称新的文献优先权或目标行列式。

Keywords: Lyness map; elliptic oval; Abel rotation; rational torsion; return shear; orbit counting
关键词：莱尼斯映射；椭圆卵形线；阿贝尔旋转；有理扭点；返回剪切；轨道计数

1 One source, two periodicity questions

For $a > 0$ consider the autonomous map and its invariant

$$F_a(x,y) = \left(y, \frac{a+y}{x}\right), \quad Q = (0,\infty)^2, \quad H_a(x,y) = \frac{(x+1)(y+1)(x+y+a)}{xy}. \tag{1}$$

A least period is a positive iterate count. In particular, the historical term “prime period” can mean least period; here a *prime integer* always means an integer prime. The rational subset is considered only when $a \in \mathbb{Q}_{>0}$; it is not interchangeable with the full real quadrant. The complete-real question asks for every point's oval, rotation, repetitions and return derivative. The arithmetic question asks which of those periodic families actually contain positive rational points.

Ownership and local progress. The autonomous Lyness family has strong classical owners. Bastien–Rogalski give its global real behavior and rotation endpoints [1]. Beukers–Cushman prove an autonomous monotonicity theorem [2]; this paper does not need global monotonicity and does not turn it into a claim of a nonzero derivative everywhere. Gasull–Mañosa–Xarles study rational periods and the elliptic normal form [3]. The deep rational torsion input is Mazur's theorem [4]. The repository predecessor C173 treated only the identity $F_1^5 = I$. Here the actual step is the whole positive-real foliation and its arithmetic boundary, not another finite prefix at the same parameter. All literature-dependent conclusions below are labeled; no priority claim or target spectral identification is made.

2 Every positive orbit is accounted for

Let $\ell = (1+\sqrt{1+4a})/2$, $c_a = (\ell,\ell)$ and $h_* = (\ell+1)^3/\ell$. Put $\omega = dx\wedge dy/(xy)$ and $R(x,y) = (y,x)$. Direct substitution gives

$$F_a^{-1}(x,y) = ((a+x)/y, x), \quad RF_aR = F_a^{-1}, \quad F_a^*\omega = \omega, \quad R^*\omega = -\omega. \tag{2}$$

The Cartesian determinant is $(a+y)/x^2$, not one. Its ratio to the new coordinate product is $1/(xy)$, which proves the weighted area identity.

Theorem 1 (Global positive exhaustion)**.** *The point c_a is the unique fixed point and the unique critical point of H_a in Q. The positive fiber at $h = h_*$ is $\{c_a\}$. Every $h > h_*$ has precisely one compact analytic oval $C^+_{a,h}$. Every noncentral two-sided orbit lies on its unique such oval, bounded away from both axes.*

Proof. In $u = \log x, v = \log y$, the invariant expands to

$$K_a = e^u + e^v + e^{u-v} + e^{v-u} + (a+1)(e^{-u}+e^{-v}) + ae^{-u-v} + a + 2. \tag{3}$$

The Hessian is a sum of positive multiples of exponent-vector outer products; the vectors $(1,0)$ and $(0,1)$ already span the plane. Thus it is positive definite. The positive and negative coordinate exponentials make K_a proper. It has a unique nondegenerate minimum; symmetry and

$$\frac{d}{dr}H_a(r,r) = \frac{2(r+1)(r^2-r-a)}{r^3}$$

locate it at c_a. Along each ray from the minimum in log coordinates, strict convexity gives a strictly increasing value after the minimum, tending to infinity. Each higher level is therefore one smooth radial graph. The inverse in (2) preserves positivity, and the invariant confines every forward and backward iterate to that compact level. Solving $F_a(x,y) = (x,y)$ gives the same unique center. □

3 A normalized Abel rotation and all return matrices

Write the level equation as

$$(x+1)y^2 + B(x)y + (x+1)(x+a) = 0, \quad B(x) = (x+1)(x+a+1) - hx, \quad D(x) = B(x)^2 - 4(x+1)^2(x+a). \tag{4}$$

Let $p_h = (r,r)$ be the unique diagonal intersection with $r > \ell$. Let α,β be the minimum and maximum x coordinates on the positive oval; they are the two turning roots delimiting that oval, not arbitrarily chosen roots of the quartic. Define

$$T(h) = 2\int_\alpha^\beta \frac{dx}{\sqrt{D(x)}}, \qquad \tau(h) = 2\int_r^\beta \frac{dx}{\sqrt{D(x)}}, \qquad \rho(h) = \frac{\tau(h)}{T(h)}. \tag{5}$$

These are positive finite Abel integrals with their positive square root on the oval interval.

Theorem 2 (Complete oval dynamics)**.** *The function $\rho : (h_*,\infty) \to (0,1)$ is real analytic. There are smooth coordinates (h,θ) on every regular annulus such that*

$$F_a(h,\theta) = (h, \theta+\rho(h)), \qquad R(h,\theta) = (h,-\theta), \quad \theta \in \mathbb{R}/\mathbb{Z}. \tag{6}$$

If $\rho(h) = m/n$ in lowest terms, every point of that oval has least period n; otherwise every two-sided orbit is dense on the oval. For every positive integer N,

$$\operatorname{Fix}(F_a^N) = \{c_a\} \cup \bigcup_{\substack{h>h_* \\ N\rho(h)\in\mathbb{Z}}} C^+_{a,h}. \tag{7}$$

At a period-n point the return matrix, in these coordinates, is

$$DF_a^{kn} = \begin{pmatrix} 1 & 0 \\ kn\rho'(h) & 1 \end{pmatrix}, \qquad k \geq 1. \tag{8}$$

Its multipliers are both one. A nonzero shear is asserted only when $\rho'(h) \neq 0$.

Proof. The auxiliary Hamiltonian vector field $X_H = xy(H_y\partial_x - H_x\partial_y)$ satisfies $\iota_{X_H}\omega = dH$ and is nonzero on each compact regular oval. The map commutes with its complete oval flow because it preserves both H and ω. On the upper and lower quadratic roots in (4), respectively, $\dot{x} = \pm\sqrt{D}$. The full flow period is therefore T.

At $x = r$, the two roots are r and $(a + r)/r < r$; the latter is exactly $F_a(p_h)$'s second coordinate. The positive clockwise arc from p_h through β to that point takes time τ. Elapsed flow time from p_h, divided by T, defines θ and proves (6). The swap fixes the section and reverses the flow. Removing the square-root turning singularities by $x = \alpha + (\beta - \alpha)\sin^2 t$ proves analytic dependence on each compact regular energy interval. The turning points are simple; equivalently their affine gradient is nonzero and the log-convex oval has a nondegenerate vertical tangency. The rotation on the circle then proves the period and density statements. Iteration and differentiation of (6) give (7) and (8). □

The two-by-two Cartesian monodromy at a return point is conjugate to (8); its nilpotent part squares to zero. At the center the one-step characteristic polynomial is $\lambda^2 - \ell^{-1}\lambda + 1$, so the multipliers are $\exp(\pm 2\pi i \rho_0)$, where

$$\rho_0(a) = \frac{1}{2\pi} \arccos \frac{1}{2\ell}. \tag{9}$$

The center is Lyapunov stable by its invariant minimum, not asymptotically stable. Periodic ovals are continuous families, not isolated hyperbolic orbits. The auxiliary T and τ have not altered the source's unit-iterate clock.

4 Evidence, dependency boundaries and conclusion

The canonical finite evidence contains 24 upper-diagonal orbit controls with 288 exact map steps, ten distinct seed-induced five-cycles and one nine-cycle, totaling 59 cycle points. Five rational centers are checked separately. Five certified intervals inside the gap between classical rotation endpoints use a 128-bit rational enclosure of π and 96-bit cosine enclosures. Their 1,280 prime and composite denominator controls give 690 reduced-fraction sufficient-period witnesses. Those witnesses are rotation values in the guaranteed image, not claimed exact coordinates of reconstructed periodic ovals.

An independent checker differentiates the scalar recurrence in two dual directions, rather than importing the producer's matrix product. Symbolic checks cover the invariant, weighted symplectic law, quartic discriminant, flex, exceptional period and return matrices. The separate 90-digit quadratures are numerical controls, not rigorous interval certificates for Abel integrals. Universal geometry and periodicity follow from the proof and the named classical inputs; finite counts do not prove them. Repaired-hash mutations test semantic rejection as well as type and YAML closure.

The source is explicitly owner-heavy. Its real oval ledger is complete, but continuous orbit families and rational torsion do not furnish a discrete rational-prime carrier. The strict evaluator tuple is
`A0_WEAK_ARITHMETIC_RELATION`, `A1_WEAK`, `A2_FAIL`, `A3_FAIL`, `A4_FORMAL_HINT`;
the overall Route-A verdict is rejected. No target local data, Euler factors, root numbers, automorphy, target zeros, divisor, counting law, functional equation or Hilbert–Pólya operator is claimed. `NO_BAD_EULER_OR_ROOT_NUMBER`. Route B remains disabled.

References

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Round zero: complete positive ovals, Abel rotations and return matrices.